XRPL Validator — Daily Engineering Report · 2026-11-09

Period: 2026-11-09 00:00 to 24:00 US Eastern (2026-11-09 05:00 to 2026-11-10 05:00 UTC); ledgers #108,210,860 to #108,232,392. Every count from the ledger and the validator's logs stops at #108,232,392, closed 04:59:51 UTC. Generated 2026-11-10 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — 1 engine receipt open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,532 of the 21,533 ledgers the network closed (99.995%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 1 outside maintenance (0 incidents, 1 isolated miss).
- **Native engine:** 1 engine receipt: #108,219,725 — open.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 15 slow-close periods; the longest, 14:51 UTC–14:52 UTC, closed ledgers up to 30 s apart.
- **Independence program:** 12 of 30 qualifying days (snapshot at 05:20:14 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,533
Validations issued	21,532
Coverage (validations / ledgers closed)	99.995%
Validations matching the network's validated ledger hash	21,532 of 21,532
Unsigned during planned restarts	0
Unsigned outside maintenance	1
Coverage outside planned maintenance (objective 99.9%)	99.995%
Restarts	0
Basis: our own ledger hash	21,528
Basis: our own hash, witnessed by the reference node (quorum short)	4
External agreement, data.xrpl.org (UTC day 2026-11-09)	0.99985 (3 missed of 20,586)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,210,860–#108,232,392), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	1
Step 3 · order	1
Step 4 · peer data	0
Step 4 · validations	5
Step 5 · own hash	0
Writer	1

Live counters, snapshot at 05:20:14 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	110,306 ledgers, 23,523,857 transactions	110,305 fully matched; results 0, bytes 3, keys 0/0
2 · transaction records	111,675 ledgers, 23,882,425 records	23,882,225 exact; differ 196, tree 2
3 · transaction order	111,718 ledgers (mode: on)	111,714 exact; order 4, results 0, bytes 0, refused 4
4 · ledger data from peers	111,724 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	111,717 ledgers judged	111,691 agreed; differ 0, short of quorum 26
5 · our own ledger hash	111,722 ledgers	111,668 own; differ 2, partial 52
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	14:51 UTC	#108,219,724–#108,219,726 (3)	Step 4 · validations	14–18 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (2 of 3 signed)
R-02	14:51 UTC	#108,219,725	Step 2 · records, Step 3 · order, Writer	Payment tx 04482BA1E70C: AccountRoot 09A3EBD5 (modified on the network): FinalFields.Balance ours="2993253044" theirs="2993253024"	open
R-03	22:02 UTC	#108,226,134–#108,226,135 (2)	Step 4 · validations	25–26 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-09-01, INC-01 is INC-2026-11-09-01.

Incidents

No incidents.

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #108,219,725 at 14:51:31 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **12 of 30** (snapshot at 05:20:14 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-09 — tracked #108,206,424–#108,227,898	Value
Validations	21,474 (21,474 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,533 (#108,210,860–#108,232,392)
Mean close interval	4.01 s
Longest close interval	30 s (before #108,219,724)
Closes at least 15 s apart	21
Empty ledgers	0

- Slow closes 07:01:52 UTC–07:02:10 UTC: ledgers #108,212,715, 1 close at least 15 s apart, up to 18 s.
- Slow closes 13:07:11 UTC–13:08:00 UTC: ledgers #108,218,216–#108,218,217 (2), 2 closes at least 15 s apart, up to 29 s.
- Slow closes 13:08:03 UTC–13:08:20 UTC: ledgers #108,218,221, 1 close at least 15 s apart, up to 17 s.
- Slow closes 13:29:01 UTC–13:29:20 UTC: ledgers #108,218,540, 1 close at least 15 s apart, up to 19 s.
- Slow closes 14:02:00 UTC–14:02:20 UTC: ledgers #108,219,027, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:03:12 UTC–14:03:30 UTC: ledgers #108,219,042, 1 close at least 15 s apart, up to 18 s.
- Slow closes 14:51:00 UTC–14:52:40 UTC: ledgers #108,219,724–#108,219,728 (5), 4 closes at least 15 s apart, up to 30 s.
- Slow closes 14:54:01 UTC–14:54:40 UTC: ledgers #108,219,744–#108,219,746 (3), 2 closes at least 15 s apart, up to 19 s.
- Slow closes 14:54:44 UTC–14:55:00 UTC: ledgers #108,219,751, 1 close at least 15 s apart, up to 16 s.
- Slow closes 15:10:12 UTC–15:10:30 UTC: ledgers #108,219,983, 1 close at least 15 s apart, up to 18 s.
- Slow closes 18:28:01 UTC–18:28:40 UTC: ledgers #108,222,920–#108,222,922 (3), 2 closes at least 15 s apart, up to 19 s.
- Slow closes 18:28:44 UTC–18:29:00 UTC: ledgers #108,222,927, 1 close at least 15 s apart, up to 16 s.
- Slow closes 22:02:51 UTC–22:03:20 UTC: ledgers #108,226,135, 1 close at least 15 s apart, up to 29 s.
- Slow closes 23:45:41 UTC–23:46:00 UTC: ledgers #108,227,688, 1 close at least 15 s apart, up to 19 s.
- Slow closes 00:02:21 UTC–00:02:40 UTC: ledgers #108,227,937, 1 close at least 15 s apart, up to 19 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- Receipt R-02 (#108,219,725): open, under investigation.

- Unsigned #108,219,725 (14:51:31 UTC): cause not recorded.
- Committed, awaiting merge and deployment: `8ffed72` fix(transactor): AccountDelete stamps AccountTxnID before do_apply erases the root.

Corrections to earlier reports

None.